

Interview Q&A: IImS

1. What is the purpose of an activation function in a neural network?

Activation functions introduce non-linearity to the model, allowing it to learn complex relationships between inputs.

2. What is the difference between a feedforward neural network and a recurrent neural network?

A feedforward neural network is a type of neural network where data flows only in one direction, from input layer to output layer without any feedback loops. A recurrent neural network (RNN) is a type of neural network that uses feedback connections to allow information to persist over time.

3. What is the difference between a linear layer and a fully connected layer in an LLM?

A linear layer is a simple neural network layer that performs matrix multiplication, whereas a fully connected layer is a type of linear layer with additional functionality such as activation functions.

4. What is the main difference between a transformer-based LLM and a traditional RNN-based LLM?

Transformer-based LLMs use self-attention mechanisms to weigh the importance of different input elements, whereas RNN-based LLMs rely on sequential processing.

5. What is the purpose of a tokenizer in an LLM?

A tokenizer is used to split input text into individual tokens, such as words or subwords, which are then fed into the model for processing.

6. What is the role of attention mechanisms in transformer-based LLMs?

Attention mechanisms allow the model to focus on specific parts of the input sequence when computing the output, improving contextual understanding and handling long-range dependencies.

7. What is the purpose of a tokenization process in an LLM?

Tokenization breaks down input text into individual tokens, such as words or subwords, to prepare it for processing by the model.

8. What is the difference between training and fine-tuning an LLM?

Training involves pre-training a model from scratch, while fine-tuning involves adapting a pre-trained model to a specific task or dataset.

9. What is the difference between a masked language model and an unmasked language model?

A masked language model randomly masks some input tokens during training, requiring the model to predict the original token, whereas an unmasked language model sees all input tokens.

10. What is the purpose of a vocabulary in an LLM?

A vocabulary is used to map input tokens to numerical indices, enabling the model to process and understand language inputs.

11. What is the difference between a static embedding and a dynamic embedding in an LLM?

Static embeddings are pre-trained and fixed, while dynamic embeddings are learned on-the-fly for each input.

12. What is the purpose of a context window in an LLM?

A context window is used to specify the number of tokens or words that the model should consider when making predictions.

13. What is the purpose of a decoder in an LLM?

The decoder is responsible for generating output based on the input sequence, typically used in tasks such as language translation or text summarization.